



**NATIONAL
SENIOR CERTIFICATE /
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2022

**TECHNICAL MATHEMATICS P1 /
TEGNIJSE WISKUNDE V1**

MARKING GUIDELINE / NASIENRIGLYN

**MARKS/
PUNTE:** 150

Marking Codes /Nasienkodes	
A	Accuracy / Akkuraatheid
CA	Consistent Accuracy / Deurlopende akkuraatheid
M	Method / Metode
R	Rounding / Afronding
NPR	No Penalty for Rounding / Geen penalisering vir afronding
NPU	No Penalty for Units omitted / Geen penalisering vir eenhede uitgelaat
S	Simplification / Vereenvoudiging
SF	Substitution in the correct Formula / / Korrekte vervanging in die korrekte formule
AO	Answer Only / Slegs antwoord

This marking guideline consists of 14 pages. /
Hierdie nasienriglyn bestaan uit 14 bladsye.

QUESTION/VRAAG 1					
1.1					
	1.1.1	$(3 - \sqrt{x})(3 + \sqrt{x})$ $= 9 - x$	$\checkmark 9$ $\checkmark -x$	A A	(2)
	1.1.2	$\left(\frac{3}{3^{3x}} + \frac{2}{3^{3x}}\right) \div \frac{10}{27^x}$ $= \frac{5}{3^{3x}} \times \frac{27^x}{10}$ $= \frac{5}{3^{3x}} \times \frac{3^{3x}}{2.5}$ $= \frac{1}{2}$	$\checkmark \frac{5}{3^{3x}}$ $\checkmark \times \frac{27^x}{10}$ $\checkmark$ Prime factors / <i>Priemfaktore</i> $\checkmark \frac{1}{2}$	A A CA CA	(4)
	1.1.3	$\frac{\sqrt{2}(\sqrt{12} + \sqrt{75})}{\sqrt{6}}$ $= \frac{\sqrt{2}(\sqrt{2^2 \cdot 3} + \sqrt{5^2 \cdot 3})}{\sqrt{2 \cdot 3}}$ $= \frac{\sqrt{2} \cdot \sqrt{3}(2 + 5)}{\sqrt{2 \cdot 3}}$ $= 7$ OR / OF $\frac{\sqrt{2}(\sqrt{12} + \sqrt{75})}{\sqrt{6}}$ $= \frac{\sqrt{2}(\sqrt{4 \cdot 3} + \sqrt{25 \cdot 3})}{\sqrt{2 \cdot 3}}$ $= \frac{\sqrt{2}(2\sqrt{3} + 5\sqrt{3})}{\sqrt{2 \cdot 3}}$ $= \frac{\sqrt{2}(7\sqrt{3})}{\sqrt{2 \cdot 3}}$ $= 7$	$\checkmark$ Prime factors/ <i>Priemfaktore</i> $\checkmark$ Common factor / <i>Gemene faktor</i> $\checkmark$ Simplification / <i>Vereenvoudiging</i> $\checkmark$ Simplification / <i>Vereenvoudiging</i> OR/OF $\checkmark$ Prime factors / <i>Priemfaktore</i> $\checkmark$ Common Factor / <i>Gemene faktor</i> $\checkmark$ Simplification / <i>Vereenvoudiging</i> $\checkmark$ Simplification / <i>Vereenvoudiging</i>	A CA CA CA A CA CA CA	(4)

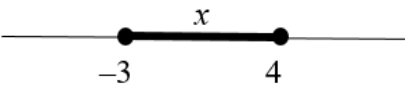
1.1.4	$\log_5 \left(\frac{1}{5} \right) + \log_5 30 - \log_5 6$ $= \log_5 5^{-1} + \log_5 \left(\frac{30}{6} \right)$ $= -\log_5 5 + \log_5 5$ $= -1 + 1$ $= 0$ OR / OF $\log_5 \left(\frac{1}{5} \right) + \log_5 30 - \log_5 6$ $= \log_5 1 - \log_5 5 + \log_5 (5 \times 6) - \log_5 6$ $= 0 - 1 + \log_5 5 + \log_5 6 - \log_5 6$ $= 0$	✓ Log property/eienskap ✓ Log property/eienskap ✓ Log property/eienskap ✓ Log property/eienskap ✓ S OR/OF ✓ Log property/eienskap ✓ Log property/eienskap ✓ Log property/eienskap ✓ Log property/eienskap ✓ S	A A CA CA CA A A A CA CA	(5)
1.2	$\frac{2(\log 1 - \log 3 - \log 2)}{\log 36} = -1$ $\text{LHS/LK} = \frac{2(0 - \log 3.2)}{\log 6^2}$ $\text{LHS/LK} = \frac{-2 \log 6}{2 \log 6}$ $\text{LHS/LK} = -1$ $= \text{RHS/RK}$	✓ Log 1 = 0 ✓ Log property/eienskap ✓ Exponential form / eksponensiële vorm ✓ Simplification / Vereenvoudiging ✓ Log property/eienskap	A A A CA CA	(5)
1.3				
1.3.1	$100111_2 \div 11_2$ $\begin{array}{r} \begin{array}{cccccc} & & & 1 & 1 & 0 & 1_2 \\ 11_2 & \overline{) 1} & 0 & 0 & 1 & 1 & 1_2 \\ & \underline{-1} & 1_2 & & & & \\ & & 0 & 1 & 1_2 & & \\ & & & \underline{-1} & 1_2 & & \\ & & & & 0 & 0 & 1_2 \\ & & & & & \underline{-0} & 0_2 \\ & & & & & & 0 & 1 & 1_2 \\ & & & & & & & \underline{-1} & 1_2 \\ & & & & & & & & 0 & 0_2 \end{array} \end{array}$ $\therefore 100111_2 \div 11_2 = 1101_2$ Remainder/Res 00_2	✓ 0 1 1₂ ✓ 0 0 1₂ ✓ 0 1 1₂ ✓ 0 0₂ ✓ 1 1 0 1₂	A CA CA CA CA	(5)

1.3.2	Binary Quotient to Decimal / Bineêre kwosiënt tot desimaal $\begin{array}{cccc} 1 & 1 & 0 & 1_2 \\ = 1 \times 2^3 + & 1 \times 2^2 + & 0 \times 2^1 + & 1 \times 2^0 \\ = 8 + & 4 + & 0 + & 1 \\ = 13 \end{array}$ OR/OF Dividend/Deeltal: $\begin{array}{cccccc} 1 & 0 & 0 & 1 & 1 & 1_2 \\ = 1 \times 2^5 + & 0 \times 2^4 + & 0 \times 2^3 + & 1 \times 2^2 + & 1 \times 2^1 + & 1 \times 2^0 \\ = 32 + & 0 + & 0 + & 4 + & 2 + & 1 \\ = 39 \end{array}$ Divisor/Deler: $\begin{array}{cc} 1 & 1_2 \\ = 1 \times 2^1 + & 1 \times 2^0 \\ = 2 + & 1 \\ = 3 \end{array}$ Division/Deelsomme $39 \div 3 = 13$	✓ M ✓ S ✓ S OR/OF ✓ M ✓ M ✓ S	A CA CA A A CA	(3)
1.4	$280\,000\,000 = 2,8 \times 10^{-7}$	✓ 2,8 ✓ 10^{-7}	A A	(2)
				[30]

QUESTION/VRAAG 2				
2.1				
2.1.1	$\frac{1}{(x)^{\frac{5}{3}}} = 32$ $(x)^{-\frac{5}{3}} = 2^5$ $(x)^{-\frac{5}{3} \times -\frac{3}{5}} = 2^{5 \times -\frac{3}{5}}$ $x = 2^{-3} = \frac{1}{8}$	$\checkmark (x)^{-\frac{5}{3}}$ $\checkmark \text{ Prime factors /}$ Priemfaktore $\checkmark \text{ Simplification /}$ Vereenvoudiging $\checkmark x = 2^{-3} / \frac{1}{8}$	A A CA CA	(4)
2.1.2	$\sqrt{x+5} - x = 3$ $\sqrt{x+5} = x+3$ $(\sqrt{x+5})^2 = (x+3)^2$ $x+5 = x^2 + 6x + 9$ $0 = x^2 + 5x + 4$ $0 = (x+4)(x+1) \text{ OR / OF } x = \frac{-5 \pm \sqrt{5^2 - 4(1)(4)}}{2(1)}$ $x \neq -4 \text{ or/of } x = -1$ $\therefore x = -1$	$\checkmark \text{ Transposing/}$ $\text{oorbring van } x$ $\checkmark \text{ Squaring both sides/}$ $\text{kwadreeer beide kante}$ $\checkmark \text{ Standard form /}$ standaardvorm $\checkmark \text{ Factors / Formula}$ Faktore / Formule $\checkmark x \neq -4 \text{ or/of } x = -1$ $\checkmark x = -1, \text{ correct value}$ /korrekte waarde Accept/aanvaar $x \neq -4$	S CA CA CA CA CA CA	(6)
2.1.3	$\log_x 3 = -1$ $x^{-1} = 3$ $\therefore x = \frac{1}{3}$	$\checkmark x^{-1} = 3$ $\checkmark x = \frac{1}{3}$	A CA	(2)

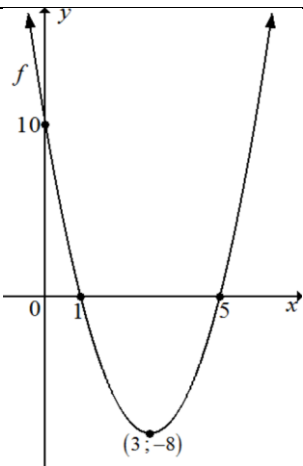
2.1.4	$\log_a (x-8) - \log_a 24 = -\log_a (x+2)$ $\log_a (x-8) = \log_a 24 - \log_a (x+2)$ $\log_a (x-8) = \log_a \left(\frac{24}{x+2} \right)$ $x-8 = \frac{24}{x+2}$ $(x-8)(x+2) = 24$ $x^2 - 6x - 16 - 24 = 0$ $x^2 - 6x - 40 = 0$ $(x-10)(x+4) = 0$ $x = 10 \text{ or } x \neq -4$ $\therefore x = 10$ OR / OF $\log_a (x-8) + \log_a (x+2) - \log_a 24 = 0$ $\log_a \frac{(x-8)(x+2)}{24} = 0$ $a^0 = \frac{(x-8)(x+2)}{24}$ $1 = \frac{(x-8)(x+2)}{24}$ $(x-8)(x+2) = 24$ $x^2 - 6x - 16 - 24 = 0$ $x^2 - 6x - 40 = 0$ $(x-10)(x+4) = 0$ $x = 10 \text{ or/of } x \neq -4$ $\therefore x = 10$	✓ Log property / eienskap ✓ Log property / eienskap ✓ S ✓ Standard form / standaardvorm ✓ Factors / Formula Faktore / Formule ✓ both values of x / beide waardes van x ✓ $x = 10$, correct value / korrekte waarde Accept/aanvaar $x \neq -4$ OR/OF ✓ Log property/eienskap ✓ Exponential form / Eksponensiële vorm ✓ Exponent 0/ Eksponent 0 ✓ Standard form / Standaardvorm ✓ Factors / Formula Faktore / Formule ✓ both values of x / Beide waardes van x ✓ $x = 10$, correct value / korrekte waarde Accept/aanvaar $x \neq -4$	A CA CA CA CA CA CA CA CA CA CA CA CA CA CA CA CA CA CA	(7)
-------	---	---	---	------------

2.2	$5 \cdot 2^{1-x} + 2^{2-x} = \frac{7}{2^{x-1}}$ $\text{LHS/LK} = 5 \cdot 2^1 \cdot 2^{-x} + 2^2 \cdot 2^{-x}$ $\text{LHS/LK} = 2^{-x} (5 \cdot 2^1 + 2^2)$ $\text{LHS/LK} = 2^{-x} (14)$ $\text{LHS/LK} = \frac{14}{2^x}$ $\text{LHS/LK} = \frac{2 \cdot 7}{2^x}$ $\therefore \text{LHS/LK} = \frac{7}{2^{x-1}} = \text{RHS/RK}$	✓ Power rule / <i>mag reël</i> ✓ Common factor / <i>gemeenskaplike faktor</i> ✓ Simplification / <i>Vereenvoudiging</i> ✓ Prime factors / <i>Priemfaktore</i>	A CA A CA	(4)
2.3				
2.3.1	$\text{Height/Hoogte (m)} = \sqrt{\frac{\text{Weight/Gewig (kg)}}{\text{BMI/LMI}}}$ $(\text{Height/Hoogte})^2 = \frac{\text{Weight/Gewig (kg)}}{\text{BMI/LMI}}$ $\therefore \text{Weight/Gewig} = \text{BMI/LMI} \times (\text{Height/Hoogte})^2 \text{ kg}$	✓ Squaring both sides / <i>Kwadreer beide kante</i> ✓ S	A CA	(2)
2.3.2	$1960 \div 1\,000 = 1,96 \text{ m}$ $\text{Weight/Gewig} = \text{BMI/LMI} \times (\text{Height/Hoogte})^2 \text{ kg}$ $\text{Weight/Gewig} = 29,9 (1,96)^2$ $\text{Weight/Gewig} = 114,86 \text{ kg}$	✓ 1,96 m ✓ Substitution / <i>Vervanging</i> ✓ 114,86 kg	A CA CA	(3)
				[28]

QUESTION/VRAAG 3				
3.1				
3.1.1	$x(x-2)-15=0$ $x^2-2x-15=0$ $(x-5)(x+3)=0$ $\therefore x=5$ or/of $x=-3$	✓ Standard form / <i>standaardvorm</i> ✓ Factors/Formula / <i>Faktore / Formule</i> ✓ 5 ✓ -3	A CA CA CA	(4)
3.1.2	$2x - \frac{7}{x} = -3$ $2x^2 - 7 = -3x$ $2x^2 + 3x - 7 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-(3) \pm \sqrt{(3)^2 - 4(2)(-7)}}{2(2)}$ $\therefore x = 1,27$ or/of $x = -2,77$	✓ Simplification <i>/Vereenvoudiging</i> ✓ Standard form / <i>Standaardvorm</i> ✓ Substitution / <i>Vervanging</i> ✓✓ Each value of x <i>Elke waarde van x</i>	A CA SF CA	(5)
3.1.3	$x^2 - x - 12 \leq 0$ $(x-4)(x+3) \leq 0$ OR / OF $x = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-12)}}{2(1)}$ $\therefore$ CVs: -3 and/en 4 $\therefore -3 \leq x \leq 4$ 	✓ Factors/ Formula <i>Faktore / Formule</i> ✓ Critical values / <i>Kritiese waardes</i> ✓ Notation / <i>Notasie</i> ✓ Number line / <i>Getallelyn</i>	A CA A CA	(4)

3.2				
	$y - x = -2$ (1) and/en $x^2 - x - 10 = y$(2) $y = x - 2$ (3) Substitute/Vervang (3) into/in (2): $x^2 - x - 10 = x - 2$ $x^2 - 2x - 8 = 0$ $(x - 4)(x + 2) = 0$ $\therefore x = 4$ or/of $x = -2$ $y = 4 - 2$ or/of $y = -2 - 2$ $y = 2$ or/of $y = -4$ OR/OF $y + 2 = x$ (3) Substitute/Vervang (3) into/in (2): $(y + 2)^2 - (y + 2) - 10 = y$ $y^2 + 4y + 4 - y - 2 - 10 - y = 0$ $y^2 + 2y - 8 = 0$ $(y - 2)(y + 4) = 0$ $\therefore y = 2$ or/of $y = -4$ $x = 4 + 2$ or/of $y = -4 + 2$ $x = 4$ or/of $x = -2$	✓ y-subject/onderwerp ✓ Substitution/ Vervanging ✓ Standard form / Standaardvorm ✓ Factors/ Formula Faktore / Formule ✓ Both x- values / Beide x-waardes ✓ Both y- values / Beide y-waardes OR/OF ✓ y-subject/onderwerp ✓ Substitution/ Vervanging ✓ Standard form/ Standaardvorm ✓ Factors/ Formula / Faktore / Formule ✓ Both y- values / Beide y-waardes ✓ Both x- values / Beide x-waardes	A CA CA CA CA CA OR/OF A CA CA CA CA CA CA	
				(6)

3.3				
3.3.1	Perimeter/Omtrek = $2l + 2b$ $2x + 2y = 18$ $\therefore x = 9 - y$	✓ Perimeter equation / <i>Omtrek vergelyking</i> ✓ Substitution / <i>Vervanging</i>	A A	(2)
3.3.2	Area = xy $\therefore \text{Area} = (9 - y)y$	✓ $(9 - y)y$	CA	(1)
3.3.3	$(9 - y)y = 18$ $9y - y^2 = 18$ $0 = y^2 - 9y + 18$ $0 = (y - 6)(y - 3)$ $\therefore y = 6$ or/of $y = 3$ $x = 9 - 6$ or /of $x = 9 - 3$ $x = 3$ or/of $x = 6$ $\therefore y = 3$ and/en $x = 6$	✓ Equation / <i>Vergelyking</i> ✓ Standard form / <i>Standaardvorm</i> ✓ Factors/ Substitution <i>Faktore/Vervanging</i> ✓ Both y values / <i>beide</i> <i>y-waardes</i> ✓ Correct values / <i>korrekte waardes</i>	A CA CA CA	(5)
3.4	$\Delta = b^2 - 4ac$ $= (1)^2 - 4(1)(1) = -3$ $\therefore$ Roots are non-real $\therefore$ Wortels is nie-reëel Accept imaginary / Aanvaar <i>imaginêr</i>	✓ SF ✓ -3 ✓ Non-real / <i>nie-reëel</i>	A CA CA	(3)
3.5	$\Delta = b^2 - 4ac = 0$ $(1)^2 - 4(1)c = 0$ $\therefore c = \frac{1}{4}$	✓ $\Delta = 0$ ✓ SF ✓ $c = \frac{1}{4}$	A A CA	(3)
				[33]

QUESTION/VRAAG 4				
4.1	$(3 ; -8)$	$\checkmark 3$ $\checkmark -8$	A A	(2)
4.2	<i>y int/afsnit:</i> $y = 2(0 - 3)^2 - 8$ $y = 10$	$\checkmark x = 0$ $\checkmark 10$		(2)
4.3	<i>x int/afsnit:</i> $0 = 2(x - 3)^2 - 8$ $4 = (x - 3)^2$ $\pm 2 = x - 3$ $\therefore x = 1 \text{ or/of } x = 5$ OR/OF $0 = 2(x^2 - 6x + 9) - 8$ $0 = 2x^2 - 12x + 10$ $0 = x^2 - 6x + 5$ $= (x - 5)(x - 1)$ $x = 5 \text{ or / of } x = 1$	$\checkmark y = 0$ $\checkmark$ Simplification / Vereenvoudiging $\checkmark$ Factors / Faktore $\checkmark$ both values of x / beide waardes van x OR/OF $\checkmark y = 0$ $\checkmark$ Simplification/ Vereenvoudiging $\checkmark$ Factors/Formule Faktore / Formule $\checkmark$ both values of x	A CA CA CA	(4)
4.4		$\checkmark$ Shape / vorm $\checkmark$ y intercept / y -afsnit $\checkmark$ Both x intercepts / beide x -afsnitte $\checkmark$ Turning point / draaipunt	A CA CA CA	(4)
4.5	$y > -8$	$\checkmark y > -8$	A	(1)
4.6	$(3 ; -6)$	$\checkmark 3$ $\checkmark -6$	A A	(2)
[15]				

QUESTION/VRAAG 5				
5.1	$m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \frac{5-0}{0-\frac{5}{3}}$ $\therefore m = -3$	✓ Formula / <i>formule</i> ✓ Substitution / <i>vervanging</i> ✓ -3	F SF CA	(3)
5.2	$c = 5$	✓ 5	A	(1)
5.3	$q = 3$	✓ ✓ 3	A	(2)
5.4	$p = 1$	✓ 1	A	(1)
5.5	$g(x) = \frac{a}{x-1} + 3$ $5 = \frac{a}{0-1} + 3$ $a = -2$	✓ Substitute/Vervang p ✓ Substitute/Vervang q ✓ Substitute point Vervang punt (5;0) ✓ value of /waarde van a	CA CA A CA	(4)
5.6	$x \in R, x \neq 1$	✓ $x \in R, x \neq 1$	A	(1)
5.7	$x \leq 5$ or/of $1 < x \leq \frac{5}{3}$	✓ $x \leq 5$ ✓ 1 ✓ $\frac{5}{3}$ ✓ correct notation / <i>korrekte notasie</i>	CA CA CA CA CA	(4)
5.8	$f(x) = \frac{2}{x-1} - 3$	✓ $\frac{2}{x-1}$ ✓ -3	CA CA	(2)
				[18]

QUESTION/VRAAG 6				
6.1	$y = -4$	$\checkmark -4$	A	(1)
6.2	At/By C: $y = 2^0 - 4$ $y = -3$ At/By D: $0 = 2^x - 4$ $2^x = 4$ $2^x = 2^2$ $\therefore x = 2$	$\checkmark x = 0$ $\checkmark y = -3$ $\checkmark y = 0$ $\checkmark$ Exponential form / <i>Eksponensiële vorm</i> $\checkmark$ Same base / <i>dieselfde</i> <i>basis</i>	A CA A CA CA	(5)
6.3	$r^2 = 2^2 = 4$ $\therefore f(x) = \sqrt{4-x^2}$	$\checkmark r^2 = 4$ $\checkmark$ Equation / <i>Vergelyking</i>	A CA	(2)
6.4	$x = 2$	$\checkmark$ value of x / <i>waarde van</i> x	CA	(1)
6.5	$EC = 2 - (-3)$ OR/OF $EC = 2 + 3$ $EC = 5$	$\checkmark$ M $\checkmark$ Length / <i>lengte</i>	A CA	(2)
6.6	The vertical line cuts the graph of f once. / <i>Die vertikale lyn sny die grafiek f eenkeer</i> OR / OF An association from the domain set to y set is many to one./ <i>'n Verbintenis vanaf die definisieversameling na die</i> <i>y-versameling is baie tot een</i>	$\checkmark$ Touch once / <i>raak</i> <i>eenkeer</i> OR / OF Many to one. / <i>baie tot een</i>	A	(1)
				[12]

QUESTION/VRAAG 7					
7.1	$i_{\text{eff}} = \left(1 + \frac{i}{m}\right)^m - 1$ $i_{\text{eff}} = \left(1 + \frac{0,063}{4}\right)^4 - 1$ Effective Interest Rate $\approx 6,45\%$ <i>Effektiewe rentekoers</i> Accept/Aanvaar $i_{\text{eff}} \approx 6,5\%$		✓ Formula / <i>Formule</i> ✓ Substitution / <i>Vervanging</i> ✓ 6,45%	F SF CA	(3)
7.2	$A = P(1+i)^n$ $8\,949 = 2\,500(1+i)^6$ $3,5796 = (1+i)^6$ $(3,579)^{\frac{1}{6}} - 1 = i$ $i = 0,2368$ $\therefore \text{Rate/Koers} = 23,68\%$		✓ F ✓ SF ✓ <i>i</i> subject/ <i>onderwerp</i> ✓ Rate as/ <i>Koers as</i> %	A CA CA CA	(4)
7.3	7.3.1	$11\% \text{ of/van } R25\,000,00 = \frac{11}{100} \times R25\,000,00$ $11\% \text{ of/van } R25\,000,00 = R2\,750,00$	✓ M ✓ R2 750,00 AO : 2 marks/punte	A A	(2)
	7.3.2	$P = R25\,000,00 - R2\,750,00 = R22\,250,00$ $A = P(1 + in)$ $A = R22\,250,00(1 + 0,16 \times 4)$ $\therefore A = R36\,490,00$	✓ $P = R2\,750,00$ ✓ SF ✓ Total Amount / <i>Totale bedrag</i>	A CA CA	(3)
	7.3.3	$\text{Monthly Repayments/} \quad = \frac{R36\,490,00}{4 \times 12}$ <i>Maandelikse terugbetalings</i> $\text{Monthly Repayments/} \quad = R760,21$ <i>Maandelikse terugbetalings</i>	✓ Method / <i>Metode</i> ✓ Answer / <i>Antwoord</i>	A CA	(2)
					[14]
TOTAL/TOTAAL:					150